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Modeling and Simulation of Speed and Efficiency of BLDC Motor as A Starter Motor Based on Multilayer Perceptron (MLP) Neural Network

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***Abstract*— This paper present the mathematical model of Permanent Magnet Brushless DC (PMBLDC) motor which can replace the current model existing in simulation tools. Currently, PMDC motor is being used as starter motor in automotive industry. PMDC motors having brushes suffers from poor efficiency and shorter life time than BLDC motors. Validation of designed BLDC motors is traditionally being performed using simulation tools like Ansys Maxwell software which is time-consuming and not efficient when parameters change. So there is a need for developing more efficient model for BLDC motors. Here we present a mathematical model obtained using Artificial Neural Network (ANN) modeling methods which resulted in considerably faster simulation time than Maxwell software. For this paper, some geometrical parameters were set as the input of the ANN. Also, efficiency and speed were set as the ANN output. The ANN model showed considerably faster evaluation time with acceptable accuracy.**

Keywords*— *Simulation; artificial neural network; multilayer percptron (MLP); BLDC motor; back propagation; starter motor

I. Introduction

Brushless permanent magnet motor operation depends on the conversion of energy from electrical to magnetic to mechanical. Because magnetic energy has an important role in producing the torque, formulation of these techniques are needed for computational purposes [1]. The most common models are based on magnetic equivalent circuit (MEC) analysis, or finite element analysis (FEA). Ansoft Maxwell software uses both techniques to simulate electrical machines. From a designer perspective, obtaining the final designed model requires substantial change of geometrical variables and therefore several simulations are needed. These repetitive processes are very costly and time-consuming.

We are proposing an artificial neural network-based (ANN-based) modeling method which resulted in much less complex model without sacrificing accuracy. Artificial neural networks are information processing systems that their designs enlivened by the studies of the ability of the human brain to learn from observations and generalize to unseen data [2]. The ability of the ANNs to be trained for any nonlinear input–output relationships based on observed data has resulted in their use in several research areas such as pattern recognition, microwave applications, control, biomedical engineering etc. [3]. In order to build a model using neural networks, they are first trained to capture the behavior of BLDC motors [4, 5]. The trained ANNs, usually called neural-network models (or simply neural models), finally are being used in high-level simulation and design, predicting accurate and fast answers according to the data they have learned from [6, 7]. In fact, ANNs provide efficient alternatives to conventional methods such as numerical modeling techniques, which have the drawback of expensive computations, or analytical methods, which could be difficult to obtain for new components, or empirical models, which have limited range and accuracy [8, 9]. This paper is organized as follows: in section II Brushless DC Motors will be presented, artificial neural network structures will be presented in section III, simulation results are shown in section IV and the last section is conclusion.

II. Brushless DC Motors

A three-phase brushless DC motor is a permanent magnet synchronous motor which has a trapezoidal back Electromotive force (EMF). For this case, the phases current are rectangular pulses as shown in Fig. 1.

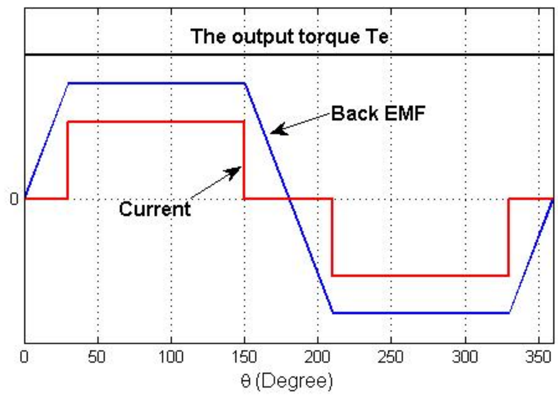

Fig. 1. Back EMF, current (phase A) and production torque

BLDC motors have many advantages to brushed DC motors including: better speed-torque characteristic, faster dynamic response, lower maintenance, longer life time, higher efficiency, lower electric noise and higher speed range. But there are some disadvantages like costly magnetic material and the probability of demagnetization. In cases that size of the motor and torque delivered ratio is higher than typical values, for example in applications where space and weight limitations exist, BLDC motors are useful [2, 3].

Currently, PMDC motor is being used as starter motor in automotive industry. BLDC motor can be good choice to replace PMDC motor as starter motor. Brushless permanent magnet motors can appear in many forms. We consider a radial flux, interior rotor with surface permanent magnet (SPM) motor which its stator is slotted as shown in the Fig. 2. The load type is constant torque (1 N.m.) and rated speed is 9000 rpm. The input voltage should be according to the typical 12-volt car battery considering voltage drop.

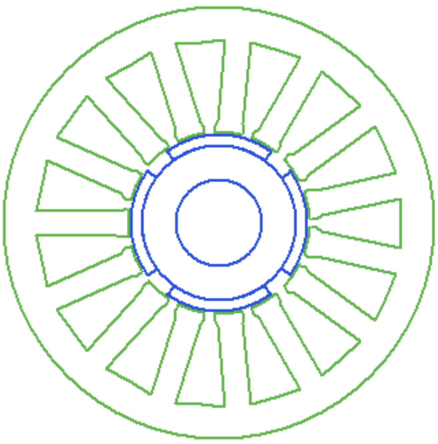

Fig. 2. Cross section view of motor

III. ARTIFICIAL NEURAL NETWORK STRUCTURES

A. Preliminaries

A basic neural network has two main elements: processing elements which are called neurons and the link between these processing elements which are associated with weights. Each neuron receives the information, processes them and then passes to its output. There are three types of neurons called input neurons, output neurons and hidden neurons. Input neurons receives the external inputs, output neurons passes the information to the final output of the model and the rest of neurons which are neither connected to the inputs nor to the final outputs are called hidden neurons.

All neurons are connected together in a layer-wise manner. Layer containing input neurons is called input layer, the one containing output neurons is called output layer and the rest of the layers are called hidden layers.

Fig. 3, shows the structure of typical multilayer perceptron (MLP) including all the elements and layers.

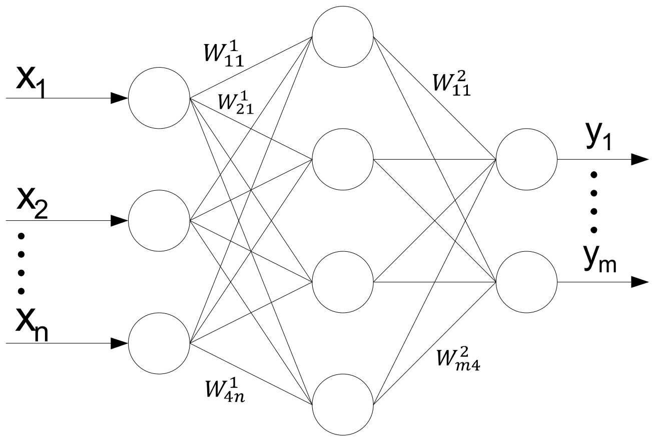

Fig. 3. A schematic diagram of a Multi-Layer Perceptron (MLP) neural network

Let *n* and *m* represents the number of input and output neurons respectively. Also, $\boldsymbol{x}$ and $\boldsymbol{y}$ are vectors containing external inputs and the outputs of the target component to be modeled and $\boldsymbol{w}$ be the vector containing all the weight parameters connecting neurons in consequent layers. The way these neurons are connected defines the structure of the neural network. Finally output of the neural network can be computed based on $\boldsymbol{x}$ and $\boldsymbol{w}$ according to the following formula:

$$\boldsymbol{y} = f_{ANN}(\boldsymbol{x}, \boldsymbol{w}) \qquad (1)$$

which f_{ANN} contains processing part (activation function) of all neurons as a nonlinear function. The original input-output relationship of the electromagnetic component can be express as:

$$\boldsymbol{y} = f(\boldsymbol{x}) \qquad (2)$$

In fact, the rule of MLP is to find an approximation (f_{ANN}) of the original relationship (f) using provided input-output training data and weights of MLP ($\boldsymbol{w}$).

B. Data Organization

The generated input-output pairs *(x, d)* could be divided into three sets, namely, training data (D_{tr}), validation data (D_v), and test data (D_t). Training data is useful to adapt the weight parameters in order to minimize the training error which is the error difference between the model and the desired output. In order to check the quality of the neural network model during training process, validation data is used. Finally, for checking the trained neural network model in terms of accuracy and generalization capability, test data is applied independently. Ideally, each of the data sets D_{tr}, D_v, and D_t should represent the original component behavior $y=f(x)$.

The orders of magnitude of various input (x) and output (d) values in electrical motor application can be very different from each other. Therefore, another process is required before training and is called scaling process. Assume x, x_{min} and, x_{max} represent a generic input element in the vectors of original data. Also assume, $\tilde{x}, \tilde{x}_{min}$, and $\tilde{x}_{max}$ represent a generic element in the vectors of scaled data. Linear scaling is given by:

$$\tilde{x} = \tilde{x}_{min} + \frac{x - x_{min}}{x_{max} - x_{min}}(\tilde{x}_{max} - \tilde{x}_{min}) \quad (3)$$

and corresponding de-scaling formula is calculated as:

$$x = x_{min} + \frac{\tilde{x} - \tilde{x}_{min}}{\tilde{x}_{max} - \tilde{x}_{min}}(x_{max} - x_{min}) \quad (4)$$

Consequently, output parameters in training data, i.e., elements in $\boldsymbol{d}$, can also be scaled similarly [2].

C. Neural-Network Training

The training process is an optimization process which includes initializing weight parameters ($\boldsymbol{w}$) appropriately in order to find the optimum weights. The most commonly used procedure for MLP weight initialization is to initialize the weights with small random values e.g. in the range [-0.5, 0.5].

Training procedure is the most important step in neural model development. The training data includes pairs of $\{(x_k, d_k), and\ k \in D_{tr}\}$, which x and d are vectors of size n and m respectively representing the inputs and desired outputs of the neural network. Neural-network training error is defined as

$$E_{D_{tr}}(\boldsymbol{w}) = \frac{1}{2}\sum_{k \in D_{tr}}\sum_{j=1}^{m}\left|y_j(\boldsymbol{x_k}, \boldsymbol{w}) - d_{jk}\right|^2 \quad (5)$$

Where d_{jk} is the jth element of $\boldsymbol{d_k}$ and $y_j(\boldsymbol{x_k}, \boldsymbol{w})$is the jth neural-network output for corresponding input $\boldsymbol{x_k}$. The training goal is to adjust $\boldsymbol{w}$ such that the error function $E_{D_{tr}}(w)$ is minimized. Since $E_{D_{tr}}(w)$ is a nonlinear function of the weight parameters $\boldsymbol{w}$, iterative algorithms are usually being used in order to find the optimum weights efficiently beginning with initializing values of $\boldsymbol{w}$ elements and then iteratively update them. The back propagation (BP) training algorithm is commonly used to update $\boldsymbol{w}$ along the negative direction of gradient of training error as $\boldsymbol{w} = \boldsymbol{w} - \eta(\partial E_D/\partial \boldsymbol{w})$ which is explained with more details in [2].

IV. RESULTS

It is possible to choose many parameters of motor as inputs or outputs of the neural network. In this paper we have chosen three geometric parameters as inputs: magnet thickness, magnet embrace and axial length of stator and rotor core (Fig. 4).

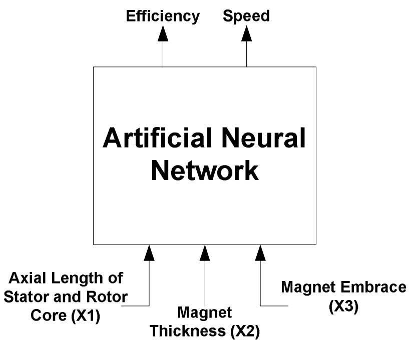

Fig. 4. A simple block diagram of ANN model including external inputs and outputs for modeling behavior of the BLDC motor

These parameters are being swiped in order to gather training data and their corresponding outputs. Also speed and efficiency of motor are selected as outputs. The training data is generated according to the following table:

TABLE I. THE INPUT RANGES OF THE NEURAL NETWORK

Geometric parameter	Unit	Minimum	Maximum
Axial Length of Stator and Rotor Core (x_1)	mm	32	38
Magnet Thickness (x_2)	mm	1	2.5
Magnet Embrace (x_3)	-	0.55	0.85

The input range in Table I is being used to generate their correspondent outputs to pass to the neural network training algorithm. In order to generate data Non-uniform Grid Distribution method is being used (each input parameter can be sampled at unequal intervals). Finally 112 training data are generated using Ansys Maxwell software. Back propagation algorithm is used to train a neural network with 1 hidden layer and 5 hidden neurons. Also *sigmoid* function is used as activation function [2]. The results of training are shown in Fig. 4, and Fig. 5:

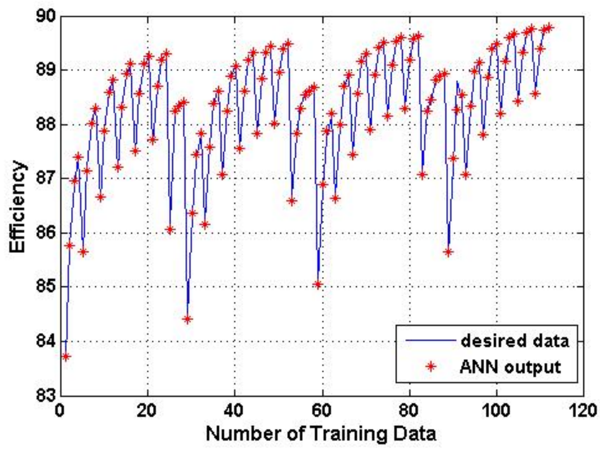

Fig. 5. Training result – Efficiency

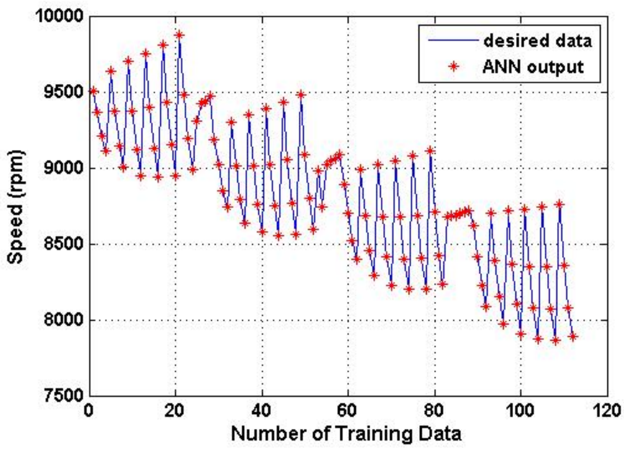

Fig. 6. Training result – Speed

In order to examine the quality of the model, some test data should be generated and applied to the obtained model to check the accuracy. The test data should not be included in training process. For this purpose 65 test data were generated in the range of training data. By applying the test data to the network these results are achieved:

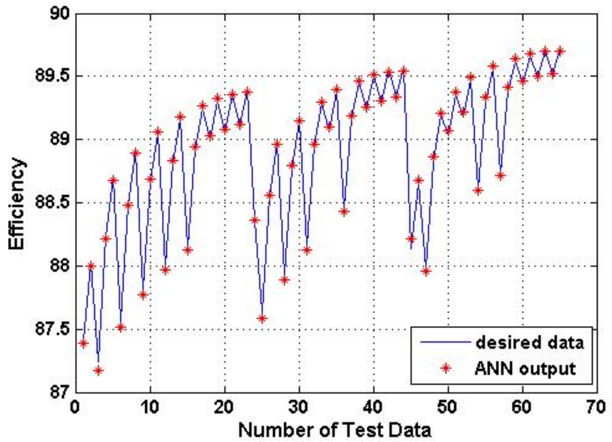

Fig. 7. Test result– Efficiency

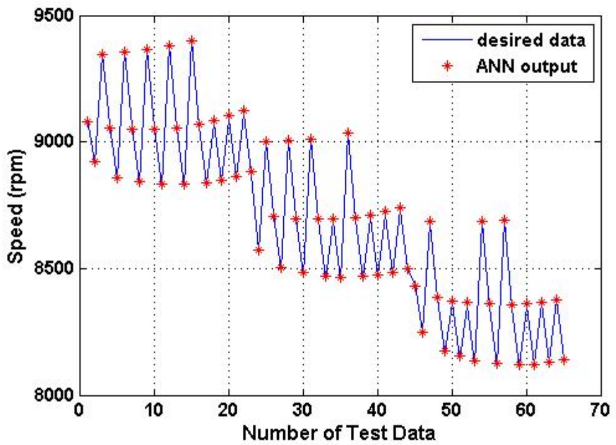

Fig. 8. Test result - Speed

Table II shows the worst case and average test error for testing the network:

TABLE II. THE WORST CASE AND AVERAGE TEST ERROR FOR ANN MODEL

Output Variable	Worst Case Test Error	Average Test Error
Efficiency	0.0071	0.000855
Speed	0.043	0.0089

Table III shows the comprising between the evaluation times of ANN model and the model used in simulation software for BLDC motor. As we expected neural network obtained model for BLDC behavior is much faster than the model being used in simulation software considering the fact that accuracy is not being sacrificed.

TABLE III. SPEEDUP COMPRISING BETWEEN ANN MODEL AND THE MODEL USED IN SIMULATION SOFTWARE FOR 65 RUN

Model type	Speedup ratio
Simulation software model	1 (reference for comprising)
ANN model	0.00026

V. CONCULUTION

In this paper, we are proposing a new method for modeling the behavior of BLDC motor based on artificial neural networks. During design process of BLDC motor, the important parameters should be changed extensively and it requires several huge CPU time and several simulations. The conventional model is being used in simulation softwares are computationally expensive. Our proposed modeling method based on neural network showed considerable speedup compare to the models used in simulators without losing accuracy. Therefore, ANN-based models can be good candidates for replacing conventional models.

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